

Questions and Answers Regarding Elementary School Solar Project: 1-31-25 (Answers at the end)

Present in person or via zoom: Doug Finn, Susan Bronstein, Todd Alexander, Scott Johndrow, Jennifer Milikowsky, Vito Perrone, Andrea McGrath, Teri Anderson. Bill Jablonski could not attend due to plowing.

1. Confirm age, size, condition, status, and remaining life of existing generators serving the property (building and well). Does the generator provide electric backup to the entire school when power is out or just critical circuits at the school? If only critical circuits, which ones?
2. Confirm current status as an emergency shelter and town's desire/plan for future use as a shelter.
 - a. According to Todd Alexander, Emergency Management Director, the Elementary school currently would be categorized as a personal care site which only provides limited services such as warming and cooling assistance, limited food and water, restrooms, and charging stations during business hours, no overnight. For overnight shelters residents would need to go to the regional shelter facility at Smith Vocational. In what situations would we use it as such? Which of these services would we provide? Who would staff it?
 - b. Is this sufficient to meet town needs? Is there a desire to designate this site as an overnight shelter or provide more services as a personal care site?
 - c. Is the current equipment satisfactory to provide the desired services in a or b? If not, what improvements would be necessary?
 - d. Is there currently air conditioning in the gym/cafeteria? If not, how would the school be used as a cooling center?
 - e. Who makes decisions regarding maintenance or repair of current equipment to ensure emergency readiness?
3. To price out a battery with microgrid SolectEnergy needs the model and serial number of the Automatic Transfer Switch (ATS) associated with the generator. Can someone provide that information?
 - a. Is there one place where all system information is kept? Unit type, installation information, maintenance logs, etc.
4. In order to appropriately size the solar array, we need to better understand current and future energy needs.
 - a. Who has confirmed the heat pumps cannot be used in winter? Have we gotten detailed information on why? With additional air sealing could they be used?
 - b. No immediate plans to make energy upgrades to the gymnasium, but is this being considered? Does the school committee consider these types of upgrades or another entity?
 - c. In order to confirm the PV system size, Solect needs to know if the school will or will not be adding mini splits or some other kind of heat pump. The current system design covers around 109% of the school's current consumption at its high points over the past 2 years. Solect can increase the system size if it's warranted.

5. Confirm final location and type of solar panels. The School Committee supported ground mounted in the location as proposed initially by SolectEnergy (in the back beyond the play area).
 - a. They also asked for Solect to evaluate the feasibility of an alternate site along the northwest side of the building near the front parking lot. During the site visit on 1/15/25, Solect did view this location and recommends keeping the original site because of its proximity and ease of access in connecting the solar panels to the electric box entering the building on the southeast side of the building. It is also a larger space and can accommodate expansion towards Hathaway Road if necessary.
 - b. Confirm a ground mounted solar panel system in the original location so that Solect can begin their system design.
6. Confirm which battery backup system is preferred and should be designed for – either a standard solar powered battery backup system or a microgrid battery backup system.
 - a. MVP grant program staff have confirmed that MVP would consider an application for a solar system that includes a standard solar powered battery backup system that is not a microgrid even though it will have limited hours of battery charge and may need to use a fossil fuel powered generator for longer power outages. The goal is to incrementally move towards a net 0 building.
 - i. MVP would not pay for any costs related to a generator and those costs would also not be allowable match, but this year MVP is likely eliminating the match requirement for small towns. (great news)
 - ii. MVP may pay for heat pumps or geothermal systems but there is no precedence. It complicates the application, but if MVP decided not to fund this portion of the project, they would remove it from the scope of work rather than deny the project (if it is a competitive project otherwise). It also could be a separate project for the future.
7. MVP would be seeking partnerships for all applications and suggests the school as a partner in student engagement. Options could include:
 - a. Student curriculum on solar energy and tracking the energy generation of the system on site.
 - b. Partner with the Grinspoon Foundation Cooler Communities Program again to assist with curriculum for this age group of students (we did this last time at the high school). The grant would pay for this assistance but teachers and curriculum development staff would need to be involved.
 - c. Partner with the COA to host one or more joint senior/student presentations on the solar system and the benefits of solar energy.
 - d. Pollinator friendly plantings in and around the solar panels – perhaps the students can be involved in planning and maintaining plantings – would need to get discuss feasibility of this with Solect from a safety and functional perspective.

Answers - Follow up questions and actions are underlined:

Generator and Oil Boiler:

- Confirmed the current generator was installed when the elementary school was constructed in 1994 – 40 years old.
- Doug Finn has paper construction plans for the school construction and roof upgrade project.
- The plan shows the location of generator and it looks like it is ahead of all wiring so that it provides power to all circuits. Andrea and Vito confirm that in Dec. when the power went out the generator came on and all systems/functions were operational.
- Bill tracks and tops off the diesel fuel when generator is running. The school oil tank is 9,000 gallons. Also, there is a backup fuel tank at the highway department with a tanker truck for transport if needed.
- Andrea did not order any oil between June 2024 and January 2025 and when it was filled, it was only filled halfway.
- Andrea has Kinsley Company conduct routine maintenance on the generator 1-2x per year. Complete overhaul servicing done in 2024.
- At that time, Kinsley recommended replacing the automatic transfer switch controller for a cost of \$6,000. Recommended because Kinsley is no longer able to get the parts to maintain the ATS controller. The town declined at the time.
- Doug has the specifications for the existing ATS and the recommended ATS controllers and will provide those electronically to Andreas at SolectEnergy.
- There is no plan by the town to replace the generator in the next couple of years. Other buildings with no generator would have higher priority for this.
- Doug recommends adding town building equipment and facilities to the capital improvements plan and there will be a discussion on this at the next CIP committee meeting.

Emergency Shelter Status:

- Todd confirms the school is a designated warming/cooling center in emergency management plan. The town does not have the facilities or staffing to operate an overnight shelter. Any overnight need would be sent to the regional Smith Vocational shelter in Northampton.
- Todd, Fire Chief, or Police Chief would call an emergency and follow procedures in emergency management plan. Incident command protocols would be initiated. To date, there has not been a need.
- Staffing of heating/cooling center would be school or town staff identified during the activation of the shelter.
- Andrea confirmed the school would not be used as a cooling/heating center when school is in session due to CORI concerns.
- There currently is no air conditioning in the gym/auditorium. Public would gather in library, classrooms, or hallways if needed as a cooling center for large numbers. Otherwise, the library would be first.

Mini-Splits:

- Currently, there are 14 mini-splits serving the elementary school installed in 2023-2024.
- As recommended by Jamrog (installers) the mini-splits are not turned on when the temperature goes below 32 degrees as they cannot keep up with the demand for heat.

- Mini-splits were not recommended for the gym because the roof structure cannot handle a roof mounted condenser, and the ceilings are too high to maintain adequate heating and cooling in the space.
- There are no immediate plans to upgrade HVAC in the gym/auditorium.
- Existing radiators in school were fitted with cold air vents during the covid pandemic for additional fresh air circulation but adds to air leakage when not used.
- The insulation that was to happen with Green Communities funding never took place.
- The elementary school electric bill is \$2500-\$3000/month during peak months.
- Phil Dowling managed the energy assessment and mini-split installation. Need to get all the information he has and document it centrally.
- Doug recommends the town consider a ground source heat pump to replace the oil boiler at some point. Would need to be part of a larger building analysis.
- Monthly measurements of oil tank to determine monthly oil usage.
- Doug recommends a building wide energy assessment to determine weatherization, insulation, equipment upgrades and to analyze whether this would allow mini-splits to function more effectively in cold temperatures.
- Doug is filing a grant application today to hire an energy manager for 3 years and will be soon applying for a \$30,000 grant to conduct a comprehensive energy assessment on the building.
- Doug will ask Floyd Fisher, Elect. Inspector, to assess the existing mini-splits - determine their current electric demand, and which circuits would be essential for a solar battery backup system.

Solar System:

- It is confirmed that the original location for the ground mount solar system as recommended by SolectEnergy is the confirmed location. Andrea and Vito in agreement. Morley says this is consistent with School Committee decision.
- A microgrid battery backup system remains the preferred solar battery backup system. The following information is needed from SolectEnergy in order to confirm:
 - Will a microgrid system allow for the removal of the oil burner and backup generator?
 - Does Solect need to install a new Automatic Transfer Switch controller for the solar system and can the generator connect to that or do they need separate AT switches?
 - Explain in detail the capacity and functionality difference between a standard solar powered energy backup system and a microgrid, especially:
 - Hours of backup power?
 - Ability to power entire school or just key circuits?
 - Do the solar panels regenerate the batteries while the batteries are being used during a power outage or are the batteries drained and then reliant on solar power or regular power or a generator to regenerate the storage?
 - Describe the structures needed for each battery backup system option and where it would be located?
 - Is it okay practice to clear snow off the panels during a power outage to ensure regeneration or is there a risk of damaging the panels?
 - Difference in maintenance between two battery systems?
 - Life expectancy of each?
 - Any safety difference?

Community Engagement/Educational Component for MVP Grant:

- Yes, the school administration is open to incorporating this if it is a requirement of the grant.
- The new curriculum director and Andrea are looking at STEM curriculum at the elementary level and this would fit that model.
- Great to tap the kid's interest during construction and in tracking the energy production.
- Yes, okay to partner with Council on Aging for joint senior/student presentations on the system.

Note from Teri: Here are notes on the battery system from questions previously asked of Solect and the answers provided. I agree further clarification is required, but wanted to remind you of this information previously provided to us.

Battery Backup Information: Andreas Response:

- Batteries must be outside usually in a shipping container like structure and includes fire detection and suppression equipment.
- Usually located at least 10 feet from the main building. Can go further, but increased cost.
- They are constructing a 100kw/204 kw-hour battery backup system at the Orange Fisher Hill Elementary School that would be about the size needed for Westhampton. Cost including prevailing wage labor is \$425,450 for the complete system install.
- The battery capacity for this size system depends on the kw usage at the school. Based on Westhampton Elementary School current peak demand in Jan 2024 at 47kw for the whole school = 4 hours of backup power. If limit backup to critical circuits like the gym, heating system, and refrigeration might get 20kw hours which = 10 hours of backup before battery is completely discharged.
- In a battery system serving critical circuits, once the battery is completely discharged, it would not come back on until power is restored.
- Would need a study of electric demand on critical circuits to determine size/cost more accurately.
- Alternative Battery System Option is a Microcircuit system which is recharged by the solar system but may be more suitable for town needs. Increased the cost for a microcircuit system is about 20-25%.
- Note that interconnecting a generator would be a more complicated design with some increased cost.